K-Means Clustering Report (Iris Dataset)

Student ID: 002362581
Student Name: Matheus Dornelles Barbosa Maia
Repository: RDatasets.jl

Section 1: Data Loading & Exploratory Analysis

Q1. Dataset overview

- Dimensions: **(150, 5)**
- Columns: SepalLength, SepalWidth, PetalLength, PetalWidth, Species
- Samples per species: setosa = 50, versicolor = 50, virginica = 50

Q2. Descriptive statistics (numeric features)

Feature	Mean	Std	Min	Max
SepalLength	5.8433	0.8281	4.3	7.9
SepalWidth	3.0573	0.4359	2.0	4.4
PetalLength	3.7580	1.7653	1.0	6.9
PetalWidth	1.1993	0.7622	0.1	2.5

Q3. Box plots by species + interpretation

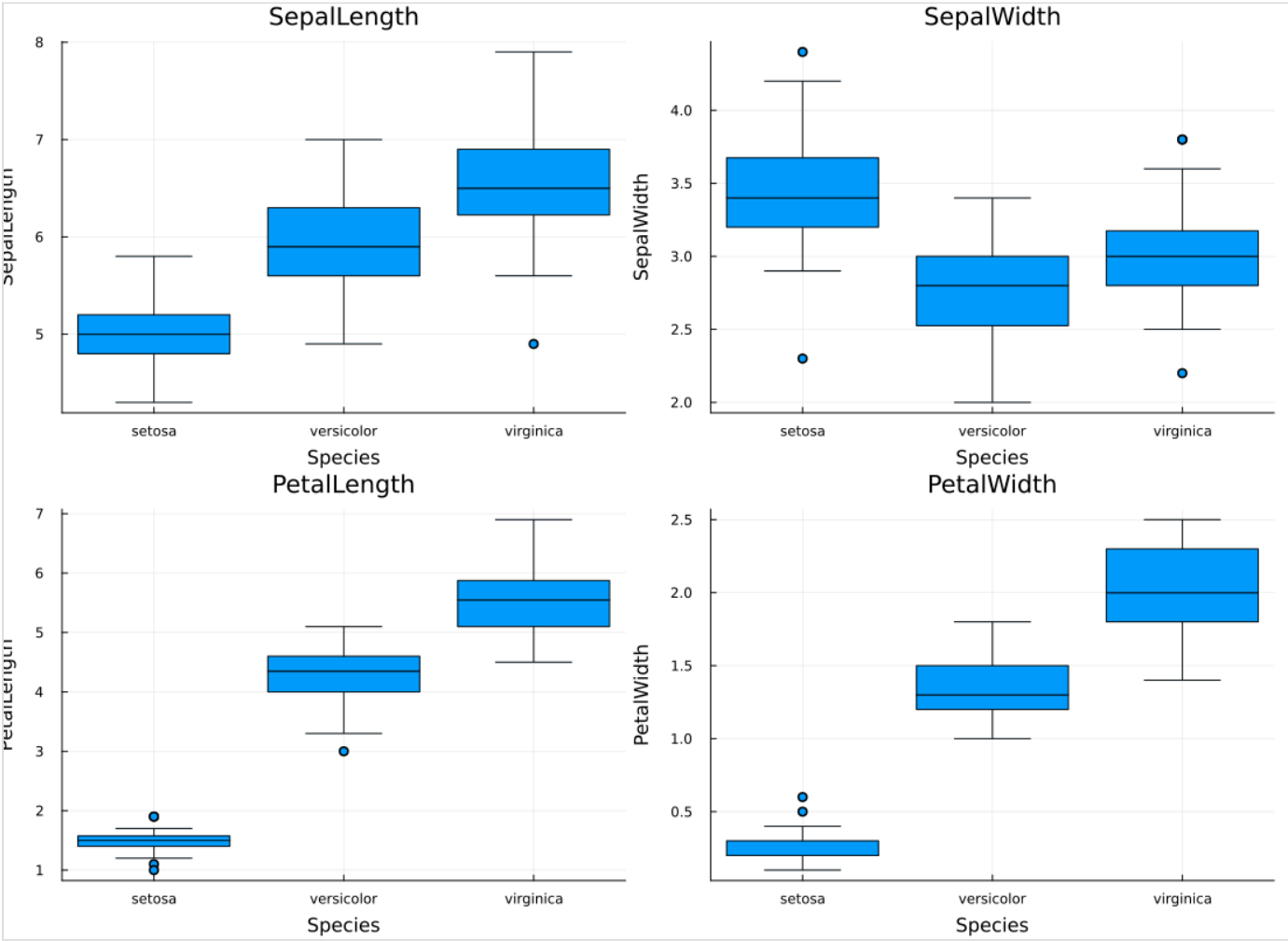

Figure Q3: Feature distributions by species.

Feature with greatest separation: PetalLength. It shows the clearest class separation with less overlap among species compared to sepal features. This is supported by its highest between-class variance proportion (eta² ≈ 0.9414), indicating strong discriminatory power.

Q4. Correlation heatmap + interpretation

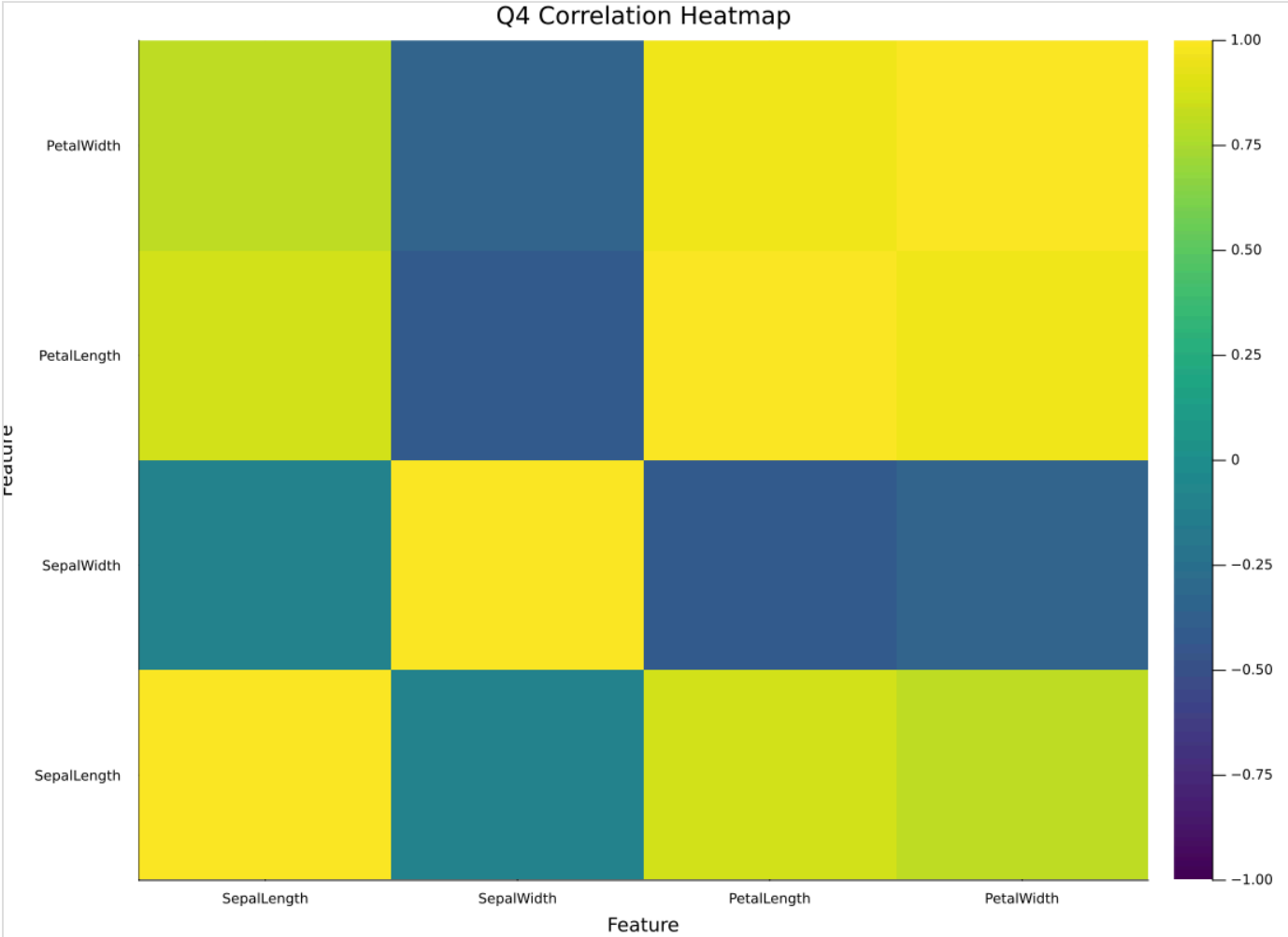

Figure Q4: Correlation matrix of numeric features.

Most strongly correlated pair: **PetalLength** and **PetalWidth** (|r| = 0.9629). This suggests feature redundancy, so dimensionality reduction (e.g., PCA) can capture most variance in fewer components with limited information loss.

Section 2: Pre-Processing & Standardisation

Q5. Matrix extraction

- Constructed numeric matrix **X** with shape **(4 × 150)**.
- Missing values check: **false** (no missing values).

Q6. Z-score standardisation verification

Feature	Mean (Before)	Std (Before)	Mean (After)	Std (After)
SepalLength	5.8433	0.8281	~0.0	1.0
SepalWidth	3.0573	0.4359	~0.0	1.0
PetalLength	3.7580	1.7653	~0.0	1.0
PetalWidth	1.1993	0.7622	~0.0	1.0

Q7. Raw vs standardised violin plots + interpretation

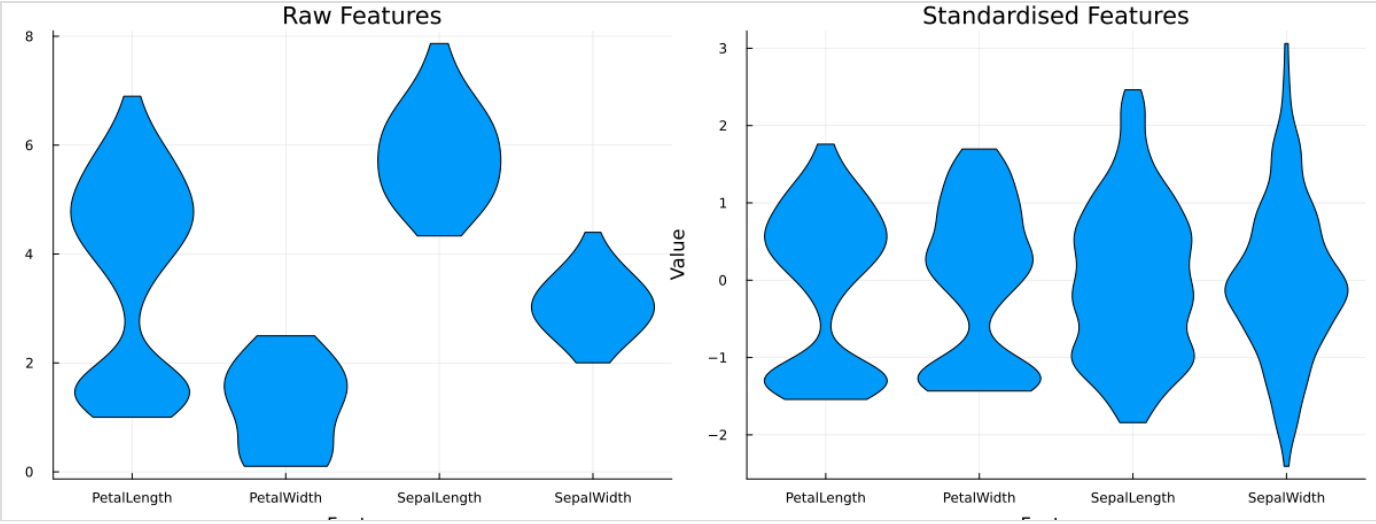

Figure Q7: Feature distributions before and after standardisation.

Raw features have different scales and dispersion, while standardised features are centered near zero with comparable spread. This matters for K-Means because Euclidean distance is scale-sensitive: without standardisation, high-variance features dominate cluster assignments.

Q8. Why omitting standardisation biases K-Means

K-Means minimises squared Euclidean distances, so features with larger variance contribute disproportionately to inertia. In the raw Iris data, standard deviations are: PetalLength = 1.7653, SepalLength = 0.8281, PetalWidth = 0.7622, SepalWidth = 0.4359. Therefore, unscaled clustering is biased toward larger-spread dimensions (especially PetalLength), reducing balanced contribution across features.

Section 3: Choosing the Optimal Number of Clusters

Q9. Elbow method (WCSS, k=1..10)

k	WCSS
1	596.0000
2	220.8793
3	139.0992
4	120.8287
5	113.1255
6	88.4012
7	80.5971
8	64.8771
9	59.4748
10	51.8204

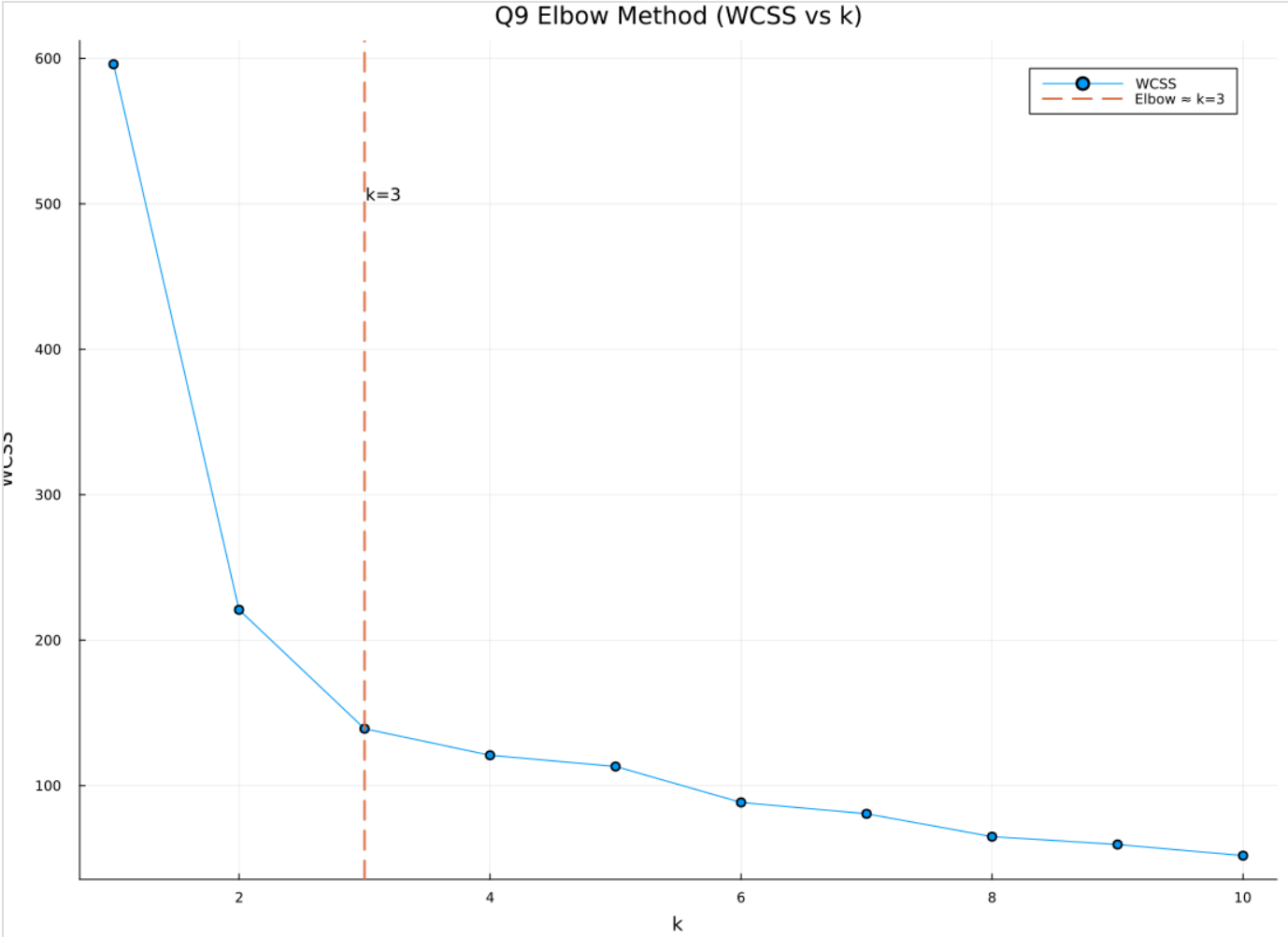

Figure Q9: WCSS vs k with elbow annotation at k=3.

Q10. Silhouette scores (k=2..10)

Rank	k	Average Silhouette
1	2	0.58175
2	3	0.46304
3	5	0.41918
4	4	0.39719
5	6	0.36162
6	8	0.35270
7	9	0.34342
8	7	0.33744
9	10	0.32966

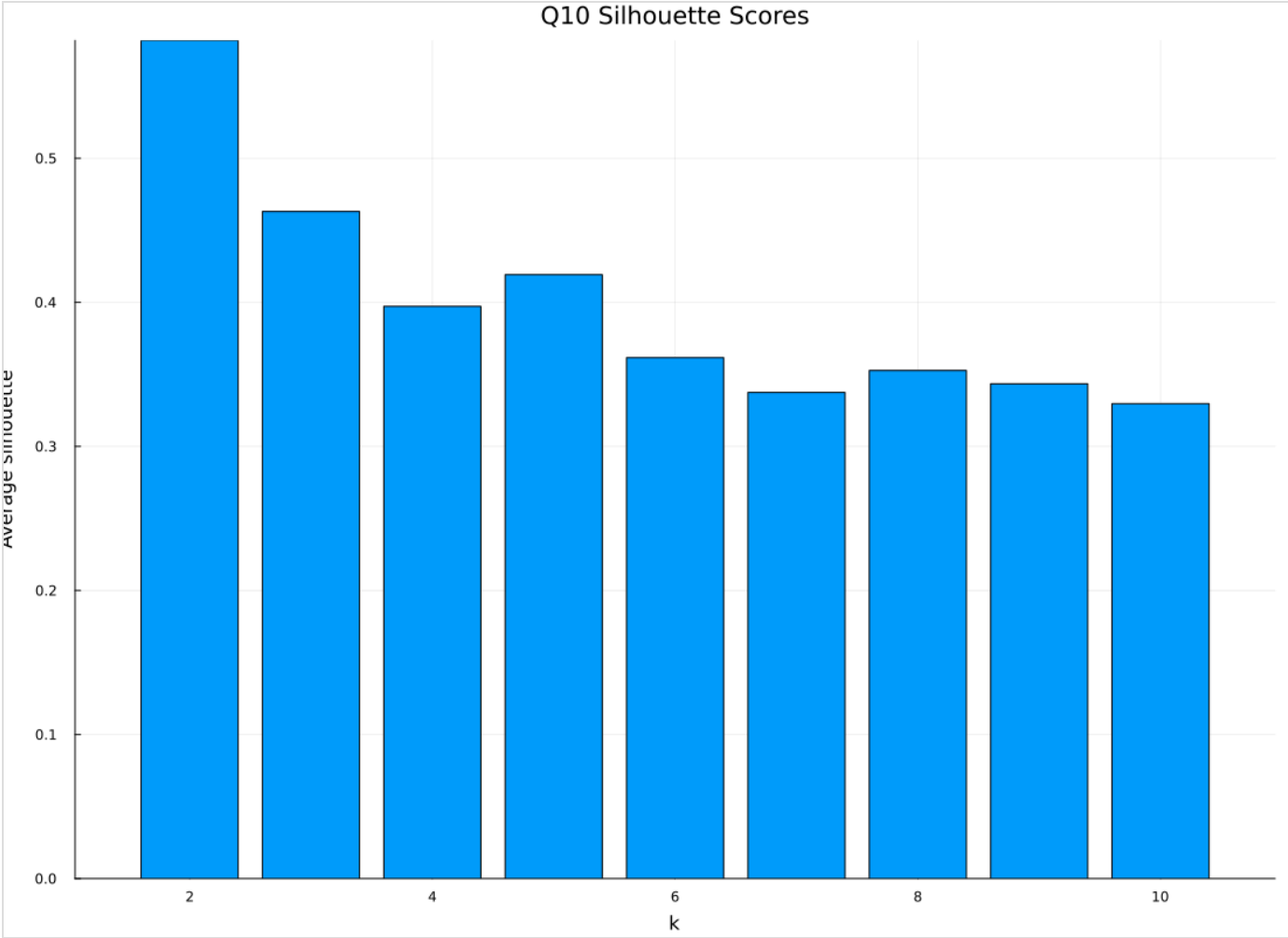

Figure Q10: Average silhouette by k.

Q11. Elbow vs silhouette (≤150 words)

Elbow analysis optimises compactness by tracking inertia (WCSS) and selecting a point of diminishing returns as k increases. Silhouette analysis optimises both cohesion and separation by comparing each sample's intra-cluster distance with its nearest neighboring cluster distance. If they conflict for Iris, silhouette is generally more informative because it directly evaluates cluster quality, not just WCSS decline. In this run, elbow indicates k≈3 while silhouette peaks at k=2, but k=3 remains a strong practical choice because Iris is known to contain three species and often one pair (versicolor/virginica) overlaps.

Q12. Why k=1 is excluded from silhouette

Silhouette requires at least two clusters. For each point, the score uses $s(i) = (b(i) - a(i)) / \max(a(i), b(i))$, where $a(i)$ is average intra-cluster distance and $b(i)$ is average distance to the nearest different cluster. With k=1, no alternative cluster exists, so $b(i)$ is undefined and silhouette cannot be meaningfully computed.

Section 4: Model Training & Cluster Profiling

Q13. Final model (k=3)

- Random seed: 42
- maxiter: 500, tol: 1e-8
- Converged: true
- Iterations: 6
- Total WCSS: 139.099201
- Cluster sizes: [56, 50, 44]

Q14. Cluster centroids

Cluster	SepalLength	SepalWidth	PetalLength	PetalWidth
Std C1	-0.0114	-0.8731	0.3758	0.3101
Std C2	-1.0112	0.8504	-1.3006	-1.2507
Std C3	1.1635	0.1448	0.9997	1.0266
Orig C1	5.8339	2.6768	4.4214	1.4357
Orig C2	5.0060	3.4280	1.4620	0.2460
Orig C3	6.8068	3.1205	5.5227	1.9818

Q15. Grouped bar chart + interpretation

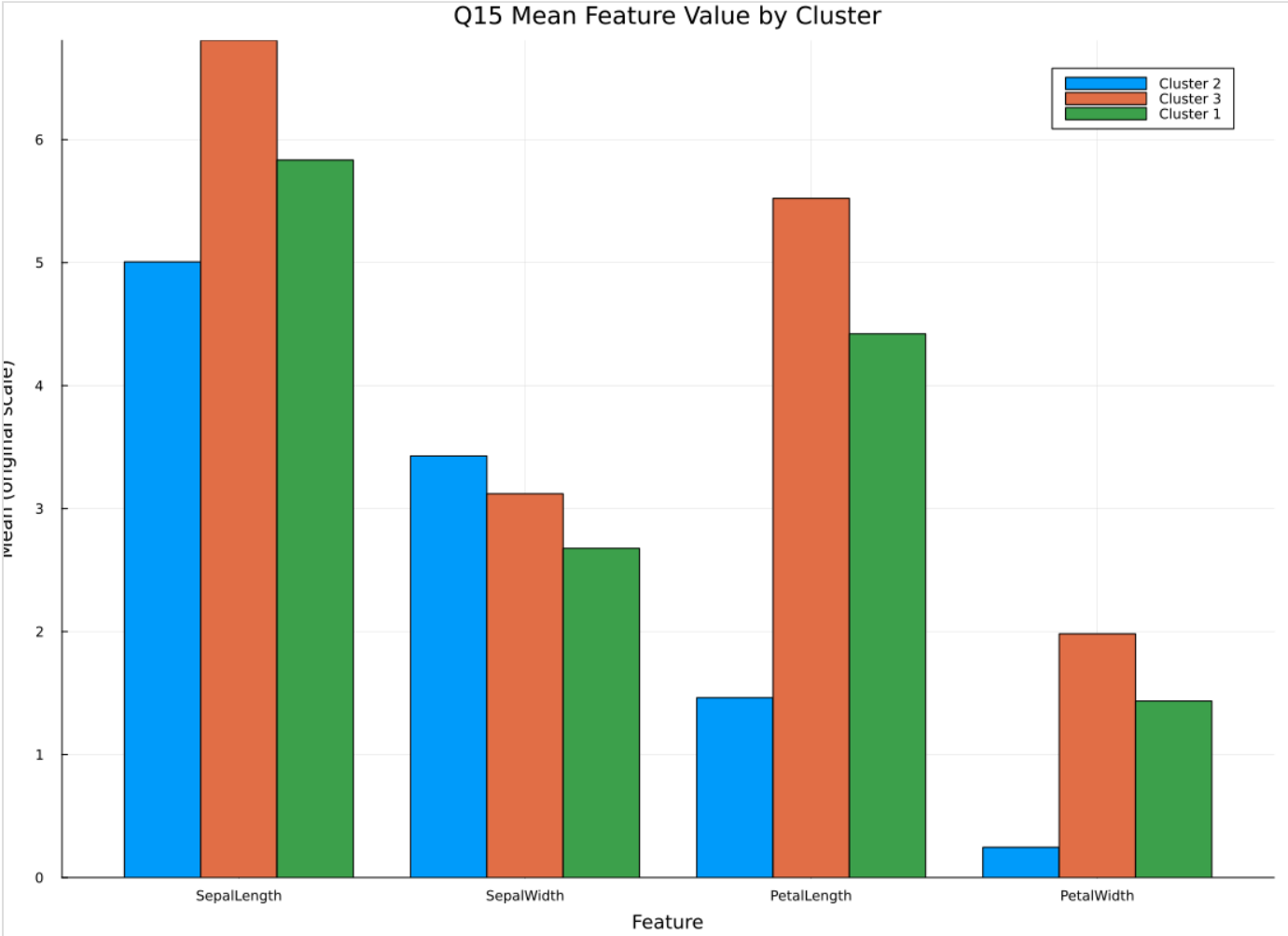

Figure Q15: Mean feature values per cluster (original scale).

Cluster 1 is most clearly distinguished by petal measurements (PetalLength and PetalWidth), which separate groups more strongly than sepal features. Sepal features vary less consistently across clusters, while petal dimensions provide clearer inter-cluster contrast.